Alexandros Tsigonias-Dimitriadis

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SUMMARY Applied scientist currently working in the Forecasting & Policy Modelling Division of DG Economics at the intersection of AI/ML and macroeconomics. I hold a Ph.D. in Mathematics, specializing in Operations Research, and have five years of research experience in algorithms, game theory and mechanism design, and mathematical optimization, alongside three years of applied experience in data science and AI/ML engineering. My research has been published in leading conferences and journals in operations research, economics and computation, and theoretical computer science.

EMPLOYMENT

European Central Bank, Germany

Economist — Sept. 2026 - present

- Working with alternative data, machine and deep learning, and advanced statistical methods to improve macroeconomic forecasting.

Graduate Programme Participant — Sept. 2024 - August 2026

- Selected as one of 15 participants from approximately 1,500 applicants for two year-long assignments and a research project, completing the first assignment in Risk Management and the second in Information Systems.
- Developed LLM-driven applications end to end as an AI/ML Engineer, including agentic and retrieval-augmented generation (RAG) tools and text classification pipelines for complex taxonomies.
- Conducted an independent project on classifying supervisory documents with limited labeled data, comparing methods for text classification, LLM-assisted annotation, and synthetic data generation.

Data Scientist — July 2023 - August 2024

- Started as a PhD trainee for one year, then transitioned into a full-time role before switching to the Graduate Programme.
- Evaluated ECB communication across general and specialized audiences using various data sources (e.g., social media, news, surveys), informing strategy through statistical analysis and custom metrics.
- Researched how media outlets portray Governing Council members' policy stances using computational social science methods.

Universidad de Chile, Chile — Aug. 2022 - July 2023

Postdoctoral Researcher, Department of Industrial Engineering

Information and Computation in Market Design (ICMD) Group (*Host*: Prof. José Correa)

- Focused on resource allocation problems arising in online markets and platforms. Topics included fairness considerations in recommendation systems, optimization aspects of political apportionment, sequential decision-making, auctions, and pricing.
- Collaborated with industry researchers, including a research visit with the Algorithms & Optimization team of Google Research Zurich.

EDUCATION

Technical University of Munich, Germany — Sept. 2018 - July 2022

Ph.D. in Mathematics, TUM School of Computation, Information and Technology

Completed an interdisciplinary PhD program with specialization in Operations Research. *Thesis*: Lotteries, Prophets, and Pandora's Box: A New Take on Classic Problems in Mechanism Design and Online Selection

Advisor: Prof. Andreas S. Schulz

National Technical University of Athens, Greece — Oct. 2012 - June 2018
Diploma (BSc & MEng) in Electrical and Computer Engineering
GPA: 8.88/10 (top 7%)
Major in Computer Science, Minor in Mathematics & Computer Networks
Thesis: Analysis of Opinion Formation Dynamics using Convex Optimization Techniques
Advisor: Prof. Dimitris Fotakis

Technical University of Munich, Germany — Apr. - Sept. 2016
Erasmus semester, Department of Electrical and Computer Engineering

RESEARCH VISITS

Google Research, Switzerland — May - July 2023
Host: Dr. Paul Dütting

Sapienza Università di Roma, Italy — Mar. 2022
Host: Prof. Stefano Leonardi

Universidad de Chile, Chile — Nov. - Dec. 2019
Host: Prof. José Correa

TECHNICAL SKILLS

Programming: Python, R, SQL, C++, MATLAB, JavaScript
Machine Learning, LLMs & Optimization: PyTorch, scikit-learn, Gurobi, NumPy, Hugging Face, LangChain/LangGraph
Software Engineering & MLOps: FastAPI, Docker, MLflow, Langfuse, Git, CI/CD, AWS, Azure

AWARDS & SCHOLARSHIPS

ICML Gold Reviewer — 2026
Top 25% of reviewers based on Area Chair ratings

COST Action STSM Grant — 2022
Funding from the COST Action for research visit at the Sapienza University of Rome

DFG-AdONE start-up funding — 2021-22
Funding from the German Research Foundation (DFG) for talented early career researchers

IGSSE PhD Scholarship — 2018-21

RESEARCH

(Authors are listed alphabetically, [C]: Conference, [J]: Journal)

A Simple Framework for Fairness of Exposure in Online Content Recommendation
José Correa, Émile Naquin, Nicolás Stier-Moses, Alexandros Tsigonias-Dimitriadis, Victor Verdugo
[C]: *Submitted* — 2026

New Combinatorial Insights for Monotone Apportionment
Javier Cembrano, José Correa, Ulrike Schmidt-Kraepelin, Alexandros Tsigonias-Dimitriadis, Victor Verdugo
[J]: **Mathematics of Operations Research** — 2025
[C]: *2025 ACM-SIAM Symposium on Discrete Algorithms* **(SODA'25)** — 2025

Prophet Secretary Against the Online Optimal
Paul Dütting, Evangelia Gergatsouli, Rojin Rezvan, Yifeng Teng, Alexandros Tsigonias-Dimitriadis
[J]: **Mathematics of Operations Research**, forthcoming — 2026
[C]: *24th ACM Conference on Economics and Computation* **(EC'23)** — 2023

Prophet Inequalities via the Expected Competitive Ratio
Tomer Ezra, Stefano Leonardi, Rebecca Reiffenhäuser, Matteo Russo, Alexandros Tsigonias-Dimitriadis
[J]: *ACM Transactions on Economics and Computation* **(ACM TEAC)** 2025
[C]: *19th Conference on Web and Internet Economics* **(WINE'23)** 2023

The Secretary Problem with Independent Sampling
José Correa, Andrés Cristi, Laurent Feuilloley,
Tim Oosterwijk, Alexandros Tsigonias-Dimitriadis
[J]: **Management Science** 2024
[C]: *2021 ACM-SIAM Symposium on Discrete Algorithms* **(SODA'21)** 2021

Preference Elicitation in Combinatorial Auctions: An Improved Approach using Reinforcement Learning
Eleni Batziou, Giorgos Bouritsas, Alexandros Tsigonias-Dimitriadis
Manuscript 2023

Robust Revenue Maximization Under Minimal Statistical Information
Yiannis Giannakopoulos, Diogo Poças, Alexandros Tsigonias-Dimitriadis
[J]: *ACM Transactions on Economics and Computation* (ACM TEAC) 2022
[C]: *16th Conference on Web and Internet Economics* **(WINE'20)** 2020

PROFESSIONAL SERVICE

Reviewer: NeurIPS ('23, '25), ICML ('23, '26), AAAI ('27), Mathematical Programming, Theoretical Computer Science
Subreviewer: COLT ('23), IPCO ('23), SODA ('23), SAGT ('22), EC ('22)
Volunteer, STOC 2022

TEACHING & SUPERVISION

European Central Bank & Barcelona School of Economics
Co-supervisor, Master thesis in Data Science Spring 2024

Technical University of Munich, Germany
Teaching Assistant, Special Topics in Algorithmic Game Theory Spring 2019
Co-supervisor of two Master theses in the Department of Mathematics

National Technical University of Athens, Greece
Teaching Assistant, Algorithms and Complexity Fall 2017

Private Tutor, Athens, Greece 2013 - 2016
Mathematics and Physics for high school students

LANGUAGES

English (fluent), Greek (native), Spanish (fluent), German (intermediate), French (intermediate)